

Prime Phase Interference Spectroscopy of Riemann Zeros: Multiscale Interaction Hierarchy, N-Body Decoupling, and Spectral Rigidity

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Abstract

We formulate an empirical prime phase interference framework that characterizes non-trivial Riemann zeros as microscopic resonance nodes along the critical line $\text{Re}(s) = 1/2$. By mapping primes to unitary logarithmic oscillators, the squared modulus of the prime Dirichlet series resolves into an invariant positive self-energy baseline $P(2) = \sum p^{-2} \approx 0.452247$ and a two-dimensional cross-coupling interaction matrix $\mathcal{M}_{pq}(t) = \frac{2}{pq} \cos(t \ln(p/q))$. Evaluating this interaction lattice across the first 200 consecutive zeros ($t_1 \approx 14.1347$ to $t_{200} \approx 396.3819$) interacting with 1,000 primes ($\binom{1000}{2} = 499,500$ pairs per zero) alongside an unconstrained random frequency baseline ($t_{\text{rand}} \in [14, 400]$) confirms that while extreme value statistics ($\cos < -0.90$ occurring at $\sim 45\%$) are globally governed by the hyperbolic amplitude envelope $2/(pq)$, individual zeros exhibit distinct microscopic phase fingerprints. Grounded in Selberg's central limit approximation, an unconstrained 1-body projection $\sum_{p \leq 1000} p^{-1/2} \cos(t \ln p)$ forms deep macroscopic resonance trenches (< -2.2) spanning the zeros, functioning as a non-perturbative suppression corridor.

Furthermore, we establish an analytical dephasing scale $\Delta t_{\text{cycle}} = \pi/|\ln(q/p)|$. While the fundamental pair $(2, 3)$ establishes the low-lying zero spacing scale ($\Delta t_1 \approx 6.89$), asymptotic coupling to high-frequency primes $q \gg p$ compresses the local dephasing scale as $\pi/\ln q$, driving the empirical 67.84% hegemony switching rate and providing a dynamical foundation for Montgomery-Odlyzko Gaussian Unitary Ensemble (GUE) level repulsion. Higher-order composite-prime harmonic couplings ($pq \leftrightarrow r$) demonstrate that composite integers act as dense secondary phase tuners, while an N -body polynomial generating function proves that multi-particle coupling magnitudes obey an exact algebraic bound $\prod p^{-1/2}$ (10^0 down to 10^{-110}). Interactions beyond $k \leq 5$ decouple naturally, capturing 99.8718% of total coupling energy and providing rigorous justification for low-order effective field theory truncations. Finally, macro-baseline calibration via the classical Riemann-Siegel formula cross-validates zero positions with a residual error of 3.2932×10^{-4} , enabling high-fidelity post-hoc spectroscopic profiling of prime phase dynamics.

Keywords: Riemann Zeta Function, Prime Phase Interference, Quantum Chaos, Random Matrix Theory, Level Repulsion, Selberg Approximation, Effective Field Theory.

*All execution scripts, numerical pipelines, raw observation datasets, and high-resolution diagnostic figures are openly maintained at: <https://github.com/CitizenKorea/k-protocol-riemann-zeros>.

1 Introduction

The spatial distribution of prime numbers and the analytic continuation of the Riemann zeta function,

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_{p \in \mathcal{P}} (1 - p^{-s})^{-1}, \quad \text{Re}(s) > 1, \quad (1)$$

form the cornerstone of modern analytic number theory [1]. Along the critical axis $\text{Re}(s) = 1/2$, the non-trivial zeros of $\zeta(s)$ exhibit deep statistical correspondences with the eigenvalue spacing distributions of random unitary matrices, known as Gaussian Unitary Ensemble (GUE) spectral rigidity [2, 3, 6]. Classical computational architectures locate zeros by evaluating the macroscopic Hardy Z -function via the Riemann-Siegel asymptotic formula [4]:

$$Z(t) = 2 \sum_{n \leq \sqrt{t/2\pi}} \frac{\cos(\theta(t) - t \ln n)}{\sqrt{n}} + \mathcal{R}(t). \quad (2)$$

While computationally fast and rigorous, integer-based asymptotic sums aggregate prime harmonics into macroscopic scalar expansions, obscuring the microscopic phase interactions occurring between individual prime degrees of freedom.

Analogous to atomic emission spectroscopy, where macroscopic absorption lines resolve into quantum state transitions, this work establishes a *Prime Phase Interference Spectroscopy*. Treating each prime as an elementary quantum harmonic oscillator with fundamental frequency $\omega_p = \ln p$ [4, 5], we construct a two-dimensional cross-coupling interaction tensor over the prime manifold $\mathcal{P}$. Rather than asserting an unsupported proof of the Riemann Hypothesis, this study formulates an empirical effective field theory model and a spectroscopic atlas of the microscopic prime resonance hierarchy governing critical zeros.

2 Spectral Invariance and Critical Axis Parameterization

The completed Riemann xi-function satisfies the functional reflection equation:

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \xi(1-s). \quad (3)$$

The algebraic symmetry under the involution $s \mapsto 1-s$ defines the coordinate invariant:

$$Q(s) \equiv s(1-s). \quad (4)$$

Parameterizing $s = \sigma + it$ in Cartesian coordinates:

$$Q(\sigma + it) = [\sigma(1-\sigma) + t^2] + it(1-2\sigma). \quad (5)$$

The quadratic invariant $Q(s)$ is strictly real-valued for non-zero vertical coordinates ($t \neq 0$) if and only if:

$$\text{Im}[Q(s)] = 0 \iff 1-2\sigma = 0 \iff \sigma = \frac{1}{2}. \quad (6)$$

Evaluating $Q(s)$ along the critical axis yields the energetic trajectory:

$$Q(s) \Big|_{\sigma=1/2} = \frac{1}{4} + t^2 \equiv K(t). \quad (7)$$

This confirms t as the natural spectral parameter governing the continuous involution symmetry along the critical line.

3 The Prime Phase Interference Matrix

3.1 Modulus Expansion of Prime Oscillators

Consider the normalized prime Dirichlet series along the boundary $\sigma = 1$:

$$P(1 + it) = \sum_{p \in \mathcal{P}} \frac{p^{-it}}{p} = \sum_{p \in \mathcal{P}} \frac{1}{p} e^{-it \ln p}. \quad (8)$$

The squared modulus $|P(1 + it)|^2$ expands into diagonal and off-diagonal pairwise products:

$$|P(1 + it)|^2 = \sum_{p \in \mathcal{P}} \sum_{q \in \mathcal{P}} \frac{1}{pq} \exp[-it(\ln p - \ln q)]. \quad (9)$$

Isolating the diagonal self-interaction ($p = q$) from off-diagonal cross-terms ($p \neq q$):

$$|P(1 + it)|^2 = \sum_{p \in \mathcal{P}} \frac{1}{p^2} + \sum_{p \neq q} \frac{1}{pq} \exp\left(-it \ln \frac{p}{q}\right). \quad (10)$$

Summing conjugate symmetric pairs (p, q) and (q, p) via Euler's identity yields:

$$|P(1 + it)|^2 = P(2) + \sum_{p < q} \frac{2}{pq} \cos\left(t \ln \frac{p}{q}\right), \quad (11)$$

where $P(2) = \sum_p p^{-2} \approx 0.452247420$ acts as an immutable positive self-energy baseline, and

$$\mathcal{M}_{pq}(t) = \frac{2}{pq} \cos\left(t \ln \frac{p}{q}\right) \equiv A_{pq} \cos(t \Delta \omega_{pq}) \quad (12)$$

defines the 2D interference matrix modulated by the beat frequency $\Delta \omega_{pq} = \ln(p/q)$ and hyperbolic amplitude envelope $A_{pq} = \frac{2}{pq}$.

3.2 Macroscopic Resonance Trenches and the Selberg Corridor

In the vicinity of a non-trivial zero $\zeta(1/2 + it_k) = 0$, regularized logarithmic expansions require destructive prime interference to counterbalance the positive ground state:

$$\Phi(t_k) \equiv \sum_{p < q} \mathcal{M}_{pq}(t_k) \ll 0. \quad (13)$$

At the macroscopic level, this destructive condensation is reflected in the 1-body projection wave:

$$\Psi_{\text{trench}}(t) \equiv \sum_{p \leq X} \frac{\cos(t \ln p)}{\sqrt{p}}. \quad (14)$$

By Selberg's central limit approximation for the Riemann zeta function [7]:

$$\ln |\zeta(1/2 + it)| \approx - \sum_{p \leq X} \frac{\cos(t \ln p)}{\sqrt{p}} + \mathcal{O}(1), \quad (15)$$

the formal divergence $\ln |\zeta(1/2 + it)| \rightarrow -\infty$ at true zeros mandates that $\Psi_{\text{trench}}(t)$ plunge into deep negative values. Rather than acting as a discrete 1-to-1 root locator, $\Psi_{\text{trench}}(t)$ forms wide macroscopic resonance trenches ($\Psi_{\text{trench}} < -2.2$) enclosing the critical zeros. Under finite prime truncations ($X = 1000$), rectangular truncation effects introduce Gibbs-like ringing (such as the local double-dip observed near t_1 at $t \approx 13.95$ and $t \approx 14.32$). Thus, the 1-body projection serves as an autonomous background suppression corridor within which exact zeros reside.

4 Extreme Value Statistics and the Null Hypothesis

To determine whether negative phase condensation is unique to Riemann zeros, we evaluate the interaction lattice against an unconstrained random frequency baseline ($t_{\text{rand}} \in [14, 400]$). For any target t , the minimal impact score across $\binom{1000}{2} = 499,500$ pairs is defined by:

$$\text{Rank}_1(t) = \min_{p < q} \left[\frac{2}{pq} \cos \left(t \ln \frac{p}{q} \right) \right]. \quad (16)$$

Because the fundamental pair $(2, 3)$ possesses $A_{2,3} = 1/3 \approx 0.3333$, while larger pairs ($p, q > 1000$) fall below 10^{-6} , extreme value selection naturally samples pairs containing small primes whose cosines drift into the anti-phase tail ($\cos \approx -1$).

Table 1: Comparative Baseline: First 200 Riemann Zeros vs. 200 Uniform Random Frequencies

Metric Parameter	200 Riemann Zeros	200 Random $t \in [14, 400]$	Physical Interpretation
Mean Impact Score $\mathcal{M}_{pq}$	-0.083585	-0.083002	Hyperbolic envelope $2/(pq)$
Median Impact Score	-0.057277	-0.056910	Central energy floor
Phase Condensation ($\cos < -0.90$)	45.20%	44.20%	Extreme value tail sampling
Severe Anti-Phase ($\cos < -0.95$)	31.90%	33.70%	Anti-phase tail saturation
Near-Perfect Anti-Phase ($\cos < -0.99$)	13.80%	13.90%	Coherent opposition limit
Rank 1 Heavyweight Share ($p, q \leq 7$)	85.50%	84.50%	Infrared potential basin
Rank 1 Pair $(2, 3)$ Dominance	37.50% (75 zeros)	38.50% (77 samples)	Primary IR energy well

Table 1 confirms that macroscopic extreme value statistics are an intrinsic geometric consequence of the hyperbolic envelope $A_{pq} = 2/(pq)$, which forms an immutable Infrared (IR) Potential Basin. The physical identity of Riemann zeros resides not in generic scalar extrema, but in their precise microscopic phase configurations and dynamical dephasing scales.

4.1 Analytical Dephasing Scales and GUE Level Repulsion

Because $\ln p$ and $\ln q$ are linearly independent over $\mathbb{Q}$, the phase angle $\theta_{pq}(t) = t \ln(p/q)$ explores $[0, 2\pi)$ ergodically. The spectral interval required for a resonant pair to rotate from full destructive opposition ($\cos = -1$) through orthogonality ($\cos = 0$) back to anti-phase ($\Delta\theta = \pi$) defines the characteristic dephasing cycle:

$$\Delta t_{\text{cycle}}(p, q) = \frac{\pi}{|\ln(q/p)|}. \quad (17)$$

For the dominant base pair $(2, 3)$, this evaluates to:

$$\Delta t_{\text{cycle}}(2, 3) = \frac{\pi}{\ln(1.5)} \approx 7.7487. \quad (18)$$

Between the first two zeros ($t_1 \approx 14.1347$ and $t_2 \approx 21.0220$), the empirical zero separation is $\Delta t = 6.8873$, matching $\Delta t_{\text{cycle}}(2, 3)$ within an 88.9% proximity. Once pair $(2, 3)$ drives destructive condensation at t_1 , it cannot immediately participate in destructive alignment at $t_1 + \epsilon$.

Along the critical line, the local mean zero spacing narrows asymptotically according to:

$$\langle \Delta t \rangle \approx \frac{2\pi}{\ln(t/2\pi)}. \quad (19)$$

At $t \sim 100$, $\langle \Delta t \rangle \approx 2.27$, which is substantially smaller than $\Delta t_{\text{cycle}}(2, 3)$. Destructive hegemony is preserved across compressed zero intervals because the active prime bandwidth expands: coupling between a base prime p and high-frequency primes $q \gg p$ yields an ultra-short dephasing cycle:

$$\Delta t_{\text{cycle}}(p, q) = \frac{\pi}{\ln(q/p)} \approx \frac{\pi}{\ln q} \rightarrow 0 \quad \text{as } q \rightarrow \infty. \quad (20)$$

These high-frequency modes compress the local dephasing scale in lockstep with the thinning zero spacing, driving the observed 67.84% hegemony switching rate between consecutive zeros and enforcing Montgomery-Odlyzko GUE level repulsion.

5 Empirical Phase Fingerprinting across 200 Zeros

Evaluating the first 200 consecutive zeros ($t_1 \approx 14.1347$ to $t_{200} \approx 396.3819$) across 1,000 primes yields 1,000 top-five profile records.

Table 2: Dominant Rank 1 Leader Hegemony across the First 200 Riemann Zeros

Rank	Prime Pair (p, q)	Zeros Captured	Share (%)	Spectral Role
1	(2, 3)	75	37.50%	Primary IR Potential Basin
2	(2, 5)	44	22.00%	Secondary IR Resonator
3	(2, 7)	27	13.50%	Tertiary IR Resonator
4	(3, 5)	15	7.50%	Mid-IR Coupler
5	(2, 11)	9	4.50%	High-Frequency Tuning Mode
6	(2, 13)	9	4.50%	High-Frequency Tuning Mode
7	(3, 7)	8	4.00%	High-Frequency Tuning Mode
Total Heavyweight Share $(p, q \leq 7)$		171	85.50%	Heavyweight Monopoly

5.1 Spectroscopic Discoveries

1. **Ground State Anchor:** Prime 2 participates in 100.0% of all zero clusters and accounts for 58.5% of all top-5 records, serving as the fundamental reference clock.
2. **Theoretical Bound Saturation:** At Zero 37 ($t_{37} \approx 116.226680$), pair (2, 3) reaches $\cos = -0.999998$, saturating the theoretical lower bound $\mathcal{M}_{2,3} \rightarrow -1/3 \approx -0.333333$. At Zero 19 ($t_{19} \approx 75.704691$), pair (5, 11) achieves floating anti-phase $\cos = -1.000000$ (impact score -0.036364).
3. **High-Frequency Inversion Anchor at t_1 :** At the fundamental zero $t_1 \approx 14.134725$, the dominant pair (2, 3) opposes cancellation ($\cos \approx +0.8515$, impact $+0.2838$). Primary destructive cancellation is assumed by pair (2, 23) ($\cos = -0.999366$, impact -0.0435). Prime 23 couples across multiple channels simultaneously: Rank 1 (2, 23) ($\cos = -0.9994$), Rank 2 (3, 23) ($\cos = -0.8696$), and Rank 5 (5, 23) ($\cos = -0.9128$), acting as a high-frequency hub locking low-frequency modes into destructive alignment.

6 Higher-Order Couplings and N-Body Effective Field Theory

6.1 Composite-Prime Harmonic Couplings ($pq \leftrightarrow r$)

To model phase interactions involving composite harmonics, we evaluate cross-couplings between quadratic composite integers pq and prime oscillators r :

$$\mathcal{M}_{pq,r}(t) = \frac{2}{pq \cdot r} \cos \left(t \ln \frac{pq}{r} \right). \quad (21)$$

At the 100th zero ($t_{100} \approx 236.524230$), while the 2-body sector is led by (2, 5) ($\cos = -0.998991$, impact -0.199798), the composite-prime spectrum reveals targeted fine-tuning modes.

Harmonics such as $21 = 3 \times 7$ and $33 = 3 \times 11$ resonate against prime 2 to attain near-perfect anti-phase alignments of -0.9955 and -0.9826 , confirming that composite numbers function as high-density secondary phase tuners.

Table 3: Top 5 Composite-Prime Harmonic Couplings at the 100th Riemann Zero ($t_{100} \approx 236.524230$)

Rank	p	q	r	Composite pq	Ratio (pq/r)	Phase Cosine	Impact Score
1	3	7	2	21	10.500000	-0.995460	-0.047403
2	3	11	2	33	16.500000	-0.982640	-0.029777
3	2	5	3	10	3.333333	-0.439107	-0.029274
4	2	5	7	10	1.428571	-0.895726	-0.025592
5	5	7	3	35	11.666667	-0.993151	-0.018917

6.2 N-Body Generating Polynomial and Algebraic Decoupling Bounds

To quantify multi-particle phase entanglement, we define the N -body generating polynomial over the first 100 primes:

$$\mathcal{P}_{100}(z; t) = \prod_{p \leq p_{100}} \left(1 + \frac{p^{-it}}{\sqrt{p}} z \right) = \sum_{k=0}^{100} S_k(t) z^k. \quad (22)$$

The maximal N -body coupling coefficient is bounded algebraically by:

$$|S_{100}(t)| = \left| \prod_{j=1}^{100} \frac{p_j^{-it}}{\sqrt{p_j}} \right| = \prod_{j=1}^{100} \frac{|p_j^{-it}|}{\sqrt{p_j}} = \prod_{j=1}^{100} \frac{1}{\sqrt{p_j}} = 1.456802 \times 10^{-110}. \quad (23)$$

Because $|p^{-it}| \equiv 1$ identically for all $t \in \mathbb{R}$, the rapid decay of multi-particle coupling magnitudes is structurally enforced by the prime harmonic denominator $\prod p^{-k/2}$.

Table 4: N -Body Coupling Energy Spectrum and Cumulative Saturation at t_{100}

Interaction Order (k)	Coupling Norm $ S_k $	Cumulative Energy (%)	Residual (%)	Dynamical Status
$k = 1$ (1-Body)	1.265627×10^0	55.6874%	44.3126%	Single-prime phase foundation
$k = 2$ (2-Body)	9.094051×10^{-1}	84.4389%	15.5611%	Pairwise cross-coupling lattice
$k = 3$ (3-Body)	5.659794×10^{-1}	95.5754%	4.4246%	Composite-prime fine tuning
$k = 4$ (4-Body)	3.192580×10^{-1}	99.1189%	0.8811%	High-order boundary closure
$k = 5$ (5-Body)	1.471650×10^{-1}	99.8718%	0.1282%	Physical Decoupling Cutoff (> 99.8%)
$k = 6$ (6-Body)	5.741716×10^{-2}	99.9864%	0.0136%	Asymptotic residual tail
$k = 7$ (7-Body)	1.899057×10^{-2}	99.9989%	0.0011%	Four-nine limit (99.99%)
$k = 10$	2.919705×10^{-4}	100.0000%	$\sim 10^{-7}\%$	Micro-perturbation limit
$k = 100$	$1.456802 \times 10^{-110}$	100.0000%	0.0000%	Algebraic extinction limit

Truncating the interaction Hamiltonian at order $k \leq 5$ captures 99.8718% of the total multi-particle coupling energy, providing mathematical justification for modeling critical zero dynamics as an effective 5-body field theory.

7 Macroscopic Baseline Calibration and Spectroscopic Profiling

To establish an unbiased benchmark, we decouple macroscopic zero location from microscopic phase analysis. Macroscopic benchmark coordinates $\tilde{t}_k$ are established via Brent's method applied to the classical Riemann-Siegel formula $Z(t)$ with first-order asymptotic remainder $\mathcal{R}(t)$.

Calibration accuracy improves systematically with increasing t due to the known asymptotic $\mathcal{O}(t^{-3/4})$ decay of the Riemann-Siegel remainder. Evaluating the microscopic tensor $\mathcal{M}_{pq}(\tilde{t}_k)$ at these calibrated anchors reproduces ground-truth leader hegemony structures with 100% fidelity.

Table 5: Riemann-Siegel Macro-Baseline Calibration across 200 Benchmark Zeros

Calibration Benchmark Metric	Numerical Benchmark Value
Mean Absolute Error (MAE Baseline)	3.293220×10^{-4}
Median Absolute Error	1.763661×10^{-4}
Minimum Absolute Error (Zero 125, $t \approx 278.475$)	1.374464×10^{-6}
Maximum Absolute Error (Zero 3, $t \approx 25.011$)	7.543928×10^{-3}
Mean Relative Calibration Error	0.000584%
Total Calibration Execution Time (200 Zeros)	27.27 seconds

8 Conclusion

The prime phase interference framework characterizes non-trivial Riemann zeros as microscopic resonance nodes on a two-dimensional interaction lattice. The destructive budget is governed by a hyperbolic envelope establishing an infrared heavyweight monopoly ($p, q \leq 7$ capturing 85.5% of zeros). While Selberg’s approximation manifests macroscopically as broad resonance suppression trenches (< -2.2), microscopic dephasing scales compress via high-frequency modes as $\Delta t_{\text{cycle}} \sim \pi / \ln q$, sustaining an empirical 67.84% hegemony switching rate and providing a dynamical mechanism for GUE level repulsion. Finally, the algebraic bound $\prod p^{-1/2}$ demonstrates that multi-particle couplings decouple exponentially beyond $k = 5$, validating an effective 5-body field theory representation of prime phase dynamics.

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